

Analysis of iPSC Gene Expression (will be changed)

Final Project

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Introduction

The discovery of human induced pluripotent stem cells (iPSCs) by Takahashi and Yamanaka in 2007 (Takahashi et al., 2007) has revolutionized the regenerative medicine field. iPSCs are created by reprogramming somatic cells into an embryonic like state using the Yamanaka transcription factors: SOX2, OCT4, KLF4, and c-Myc). These cells have vast capabilities due to their ability to model human development and diseases *in vitro*, and their potential for patient specific (autologous) and “off the shelf” (allogeneic) cell therapies (Cerneckis et al., 2024).

With that being said, the characterization of iPSC lines is essential in any experimental or manufacturing process to ensure consistency in culturing. This includes analyzing the expression of pluripotency (stem cell) markers, differentiation markers, and potential chromosomal abnormalities, to assess the population of the culture. This is commonly done using techniques such as reverse transcriptase or real time quantitative PCR (RT-qPCR) to analyze mRNA expression (Artyukhov et al., 2017), or using an antibody based experiment to analyze protein expression levels, such as Flow Cytometry or immunocytochemistry. When conducting these experiments, specifically RT-qPCR, reference genes, also known as housekeeping genes, are used to normalize the expression of other genes in a sample (Artyukhov et al., 2017). This allows the expression of genes to be compared across different samples, since it is presumed that the expression of the housekeeping gene is consistent across all of the samples. In stem cell samples, however, it has been found that the commonly used housekeeping genes, such as GAPDH, appear to fluctuate in iPSC samples (Panina et al., 2020), making it difficult to achieve accurate normalization data.

RNA Seq is a technique that sequences the entirety of RNA molecules in a given sample, rather than using gene specific primers or probes as in RT-qPCR and other techniques. This allows the entire transcriptome to be analyzed, providing a complete picture of the expression of genes

in a sample (Wang et al., 2009). Additionally, gene expression is calculated by dividing the raw counts of a gene by the total counts in a sample, eliminating the need for a housekeeping gene. This allows for an unbiased and quantitative analysis of millions of genes in sample (Wang et al., 2009). Despite these advantages, RNA-Seq is far more time consuming and expensive than RT-qPCR and other experiments, making it non practical in a manufacturing setting. Therefore, utilizing RNA-Seq to analyze iPSC samples could provide insight into housekeeping gene expression, allowing researchers to select ideal reference genes for their analysis.

A similar study, conducted by Carcamo-Orive *et al.*, investigated the variability of gene expression across 317 iPSC lines generated from 101 individuals (Carcamo-Orive et al., 2017) and created a publically available data set of their raw RNA Seq data. They found that genetic variability in the iPSC lines was influenced by a donor specific gene expression pattern, specifically influenced by donor age, sex, ancestry, and reprogramming variabilities. iPSCs can also be reprogrammed from a variety of somatic cells, such as blood, skin fibroblasts, hair keratinocytes, urine cells, mesencymal stem cells, and bone marrow (Poetsch et al., 2022). While some somatic cells are easier to obtain than others, such as blood cells and fibroblasts, iPSCs from these lines are associated with higher genomic mutations due to the somatic cells' high turnover rate. Additionally, iPSCs appear to have an epigenetic memory of their tissue of origin, which has been found to influence their differentiation (Poetsch et al., 2022). All of these factors, and many others, could be contributing to the variations in gene expression in iPSC samples.

This publicly available dataset from Carcamo-Orive *et al.* was used to analyze the expression of RNA across iPSC samples, with a focus on the expression of pluripotency markers, differentiation markers, and housekeeping genes.

Methods

Data Cleaning

RNA-Seq counts per million data was acquired from Stemformatics using the Carcamo-Orive (2017) dataset [Carcamo-Orive et al. (2017)]. Genes in this study were referred to by their Ensembl ID, so gene information, such as name and RNA type, was integrated from the Ensembl database.

further cleaning of the count per million data will be added

Figure 1

Genes detected in the RNA Seq counts per million data were grouped into the following categories: Protein coding, long non coding RNA (lncRNA), Small RNAs, Pseudogenes, and

Other. A count of the number of genes in each category was then performed by using the dplyr package (Wickham et al., 2023) from tidyverse (Wickham, 2023). A fraction and percentage were then calculated by dividing the count by the sum of count (i.e. total count for all categories). The scales (Wickham, Pedersen, et al., 2025) package from ggplot2 (Wickham, Chang, et al., 2025) was used to create a pie chart of the counts by RNA category.

continue methods for following figures ...

Results



Figure 1: Detected RNA Types in iPSC Sample Analyzed using RNA Seq. The number of genes detected in each RNA category from RNA-seq data were counted and plotted as a pie chart (left). Small categories ($< 2\%$), such as rRNA, tRNA, and immunoglobulin genes, were combined into an ‘Other’ category. The expression of genes, using raw count per million data, was averaged by type of RNA and plotted as a pie chart (right).

need to fix labels so they line up with slices

The RNA Seq data is composed of a variety of different types of RNA, as seen in Figure 1. 34.8% of read genes are mRNA protein coding genes, whereas the remaining 65.2% of the detected genes are not protein coding and are primary composed of noncoding RNAs and pseudogenes, which are RNA sequences that look like protein coding genes but are not able to be translated into a protein. 25.6% of the detected genes were pseudogenes. 25.5% were long non coding RNA sequences and 8.5% were small RNAs, which included snoRNA, snRNA,

scaRNA, sRNA, and ribozyme RNA. 5.7% were “Other” RNAs, which include ribosomal RNA, immunoglobulin genes, T-cell receptor genes, and mitochondrial RNAs. The right pie chart, on the other hand, displays the expression of these groups. For each gene for each sample, a count per million number was provided by Carcamo-Orive *et al.*. This is typically calculated by dividing the raw count of the gene by the total reads in the sample and multiplying by one million. The total expression was summed for each category, and this was graphed in Figure 2, which displays that 91.7% of the expression is from protein coding genes. The remaining 8.3% accounts for the expression of small RNAs, long non coding RNA, pseudogenes, and “Other” genes. This is supported by _____ data from _____ article, which saw _____.

Figure X. Will potentially add a PCA plot to identify which variables strongly influence gene expression.

Analysis of Variance Table

Response: PC1

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
sex	1	3033	3033.4	0.7567	0.385
Residuals	306	1226658	4008.7		

Analysis of Variance Table

Response: PC2

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
sex	1	713	713.05	0.2855	0.5935
Residuals	306	764263	2497.59		

Analysis of Variance Table

Response: PC1

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
age_decade	3	21643	7214.5	1.8155	0.1443
Residuals	304	1208048	3973.8		

Analysis of Variance Table

Response: PC2

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
age_decade	3	9096	3032.1	1.2194	0.3027
Residuals	304	755880	2486.4		

Analysis of Variance Table

Response: PC1

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
age_sex	7	29038	4148.3	1.0365	0.4055
Residuals	300	1200653	4002.2		

Analysis of Variance Table

Response: PC2

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
age_sex	7	71066	10152	4.3891	0.0001177 ***
Residuals	300	693911	2313		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Analysis of Variance Table

Response: PC1

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
race	4	114273	28568.3	7.6996	6.617e-06 ***
Residuals	287	1064872	3710.4		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Analysis of Variance Table

Response: PC2

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
race	4	94361	23590.3	10.251	9.011e-08 ***
Residuals	287	660466	2301.3		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

top 20 genes:

sex linked RPS4Y1 Ribosomal protein S4, Y-linked. Present only in males. High variance indicates sex is a strong driver. XIST X-inactive specific transcript. Expressed in females for X-chromosome inactivation. Sex-linked. DDX3Y DEAD-box helicase, Y-linked. Male-specific. TSIX Antisense of XIST, regulates X-chromosome inactivation. Female-specific. TXLNGY Y-linked, involved in transcription/translation regulation. KDM5D Lysine demethylase 5D, Y-linked. Male-specific. USP9Y Ubiquitin-specific peptidase 9, Y-linked. Male-specific. EIF1AY

PCA – Biplot

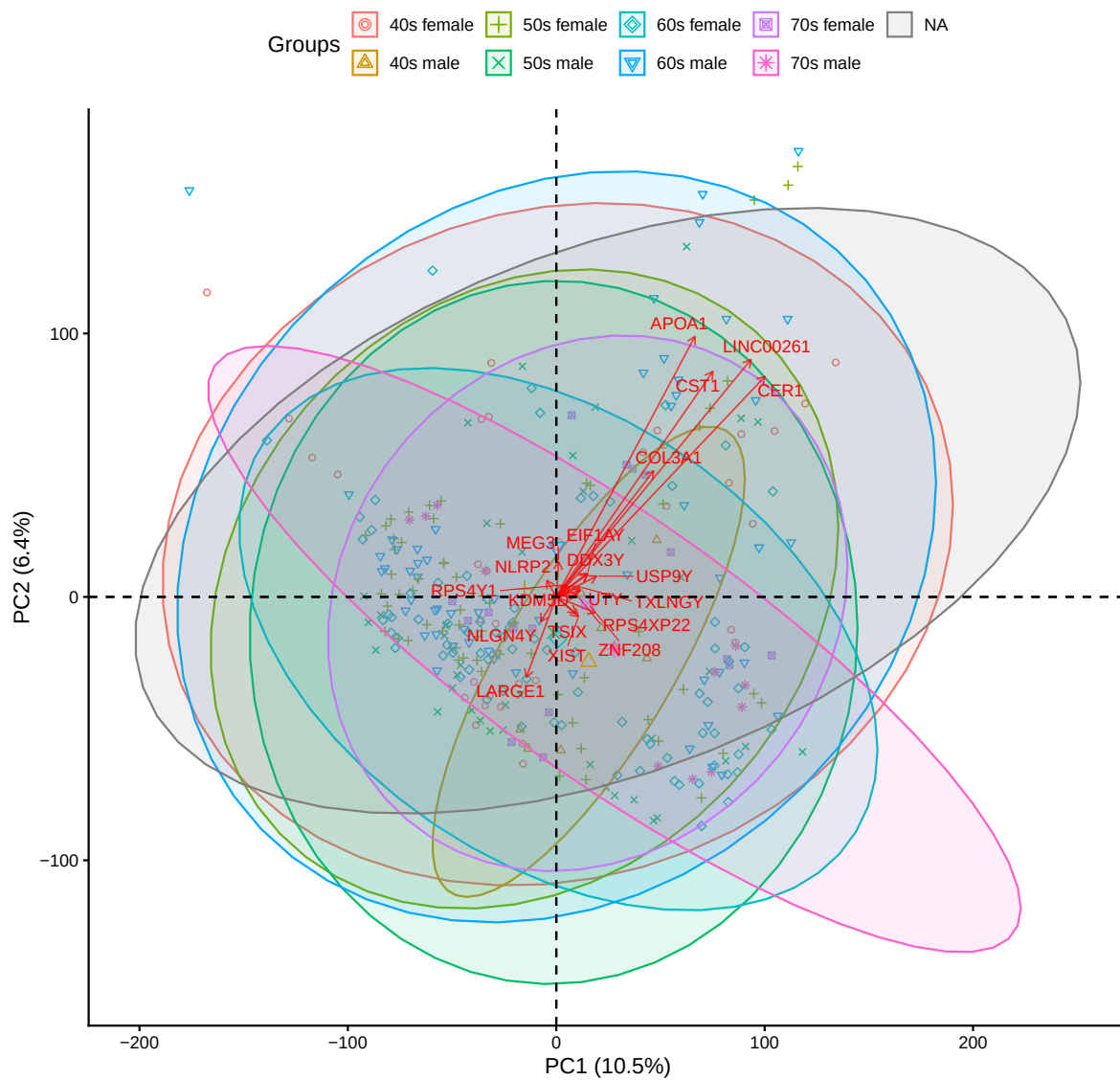


Figure 2: Caption.

PCA – Biplot

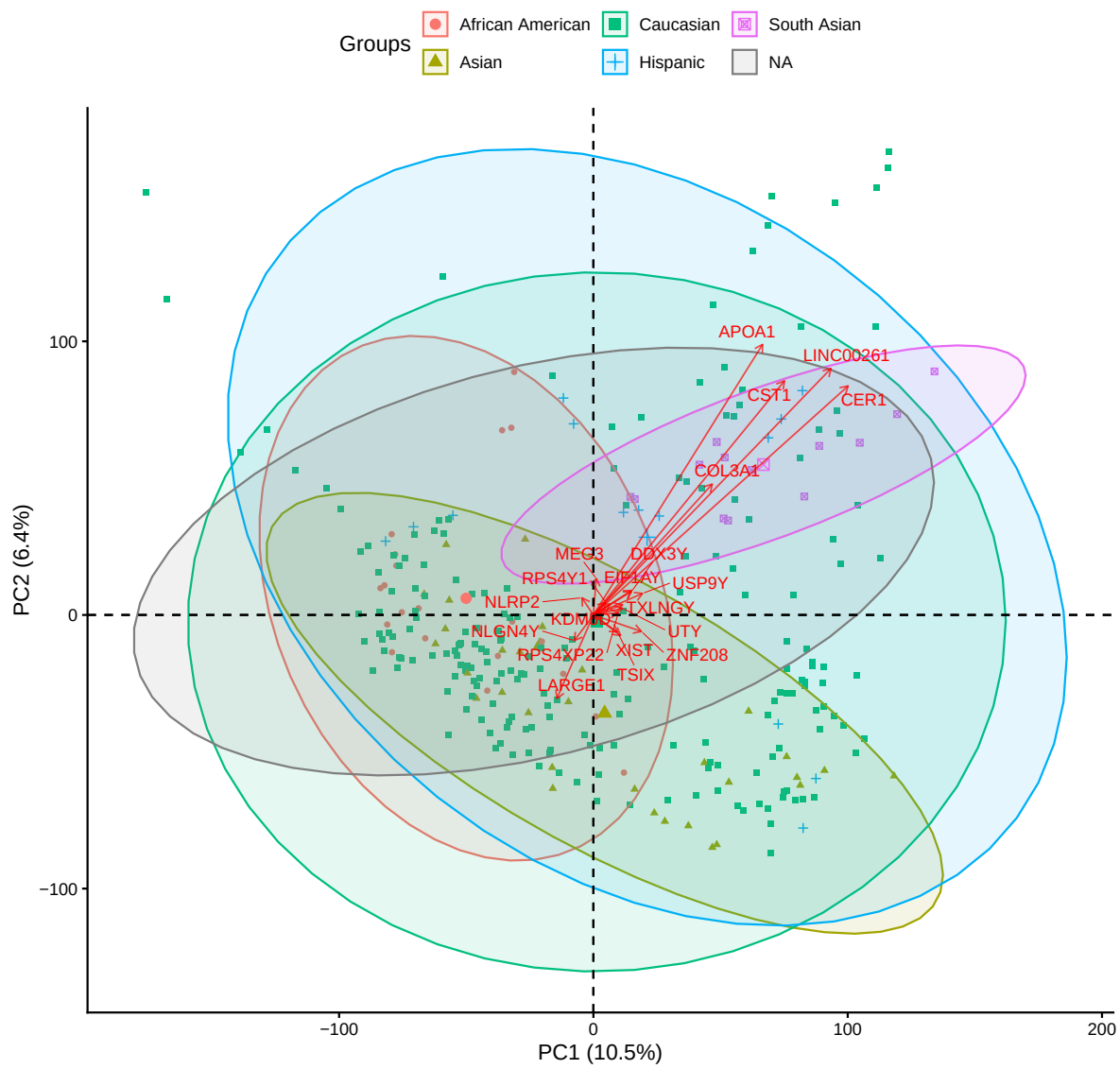


Figure 3: Caption.

Eukaryotic translation initiation factor 1A, Y-linked. Male-specific. UTY Ubiquitously transcribed tetratricopeptide repeat, Y-linked. Male-specific. NLGN4Y Neuroligin 4, Y-linked. Male-specific. RPS4XP22 Pseudogene similar to RPS4X/RPS4Y.

other genes CST1 Cystatin SN, extracellular protease inhibitor. May vary in differentiation. CER1 Cerberus, secreted factor involved in early development and differentiation. LARGE1 Glycosyltransferase, involved in post-translational modification. NLRP2 NLR family pyrin domain containing 2, involved in early development, oocyte/embryo regulation. COL3A1 Collagen type III alpha 1, ECM protein, differentiation-related. ZNF208 Zinc finger protein, transcription factor, potential regulatory role. APOA1 Apolipoprotein A1, lipid transport, may reflect metabolic differences. MEG3 Long non-coding RNA, tumor suppressor, epigenetic regulation, differentiation-related. LINC00261 Long non-coding RNA, involved in endoderm differentiation.

Pluripotency Markers

Expression of Pluripotency Genes (SOX2, POU5F1(OCT4), KLF4, MYC (c-Myc), TRA-1-81 and 1-60 (PODXL), NANOG, FUT4 (SEA4), LIN28A, LIN28B, ESRRB, PRDM14, ZRP42 (REX1), TFCP2L1.

will remove lmm and qqplot outputs for final submission

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: Expression ~ age_sex + (1 | `Gene name`)
Data: df_pluri_agesex
```

REML criterion at convergence: 4575.3

Scaled residuals:

Min	1Q	Median	3Q	Max
-4.6491	-0.5579	0.0284	0.5521	5.6655

Random effects:

Groups	Name	Variance	Std.Dev.
Gene name	(Intercept)	12.0051	3.4648
Residual		0.2074	0.4554

Number of obs: 3504, groups: Gene name, 12

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	6.19511	1.00049	11.01081	6.192	6.76e-05 ***
age_sex40s male	0.11258	0.05213	3485.00000	2.159	0.0309 *

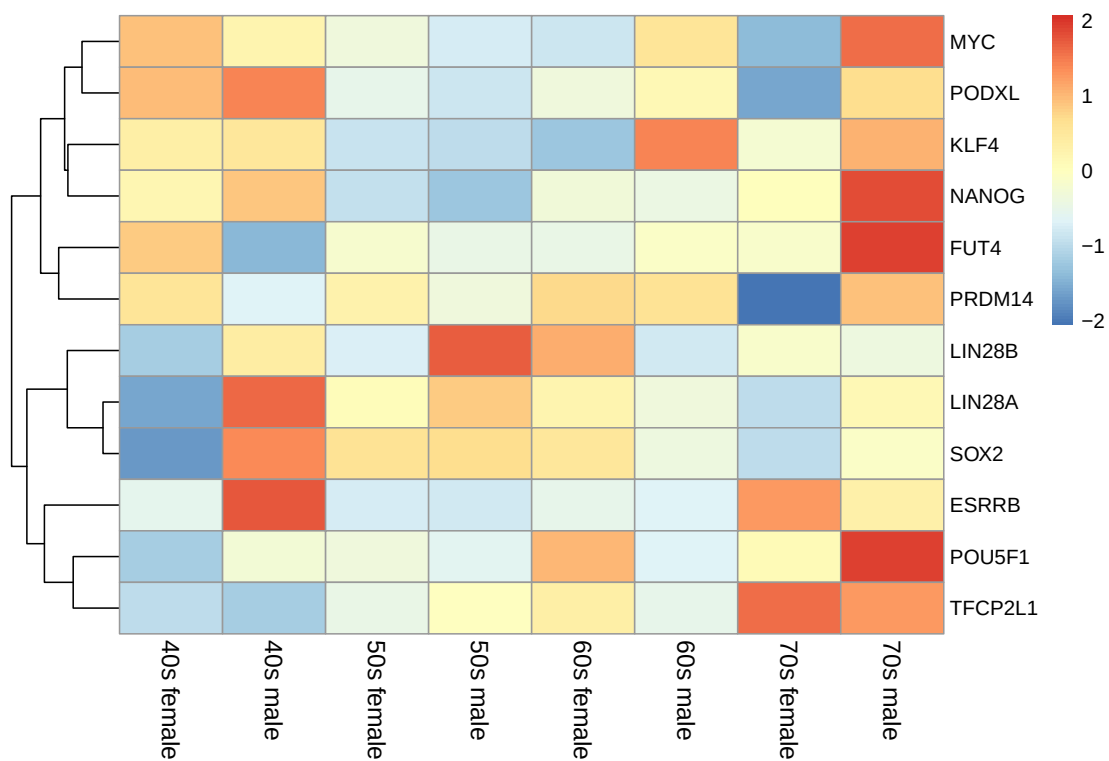


Figure 4: Expression of Pluripotency Genes

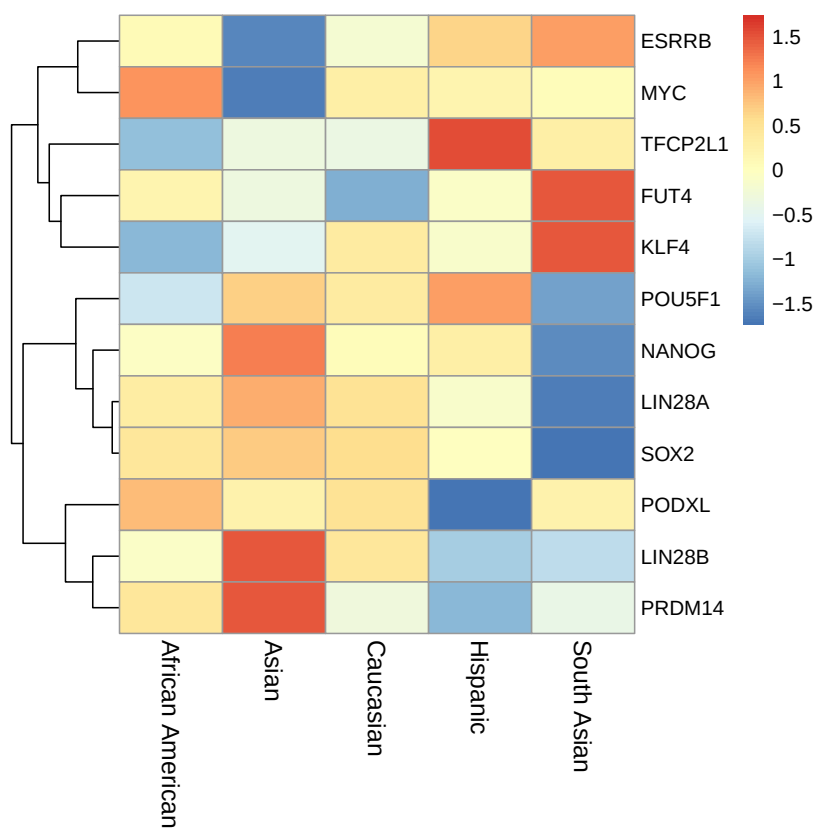


Figure 5: Expression of Pluripotency Genes

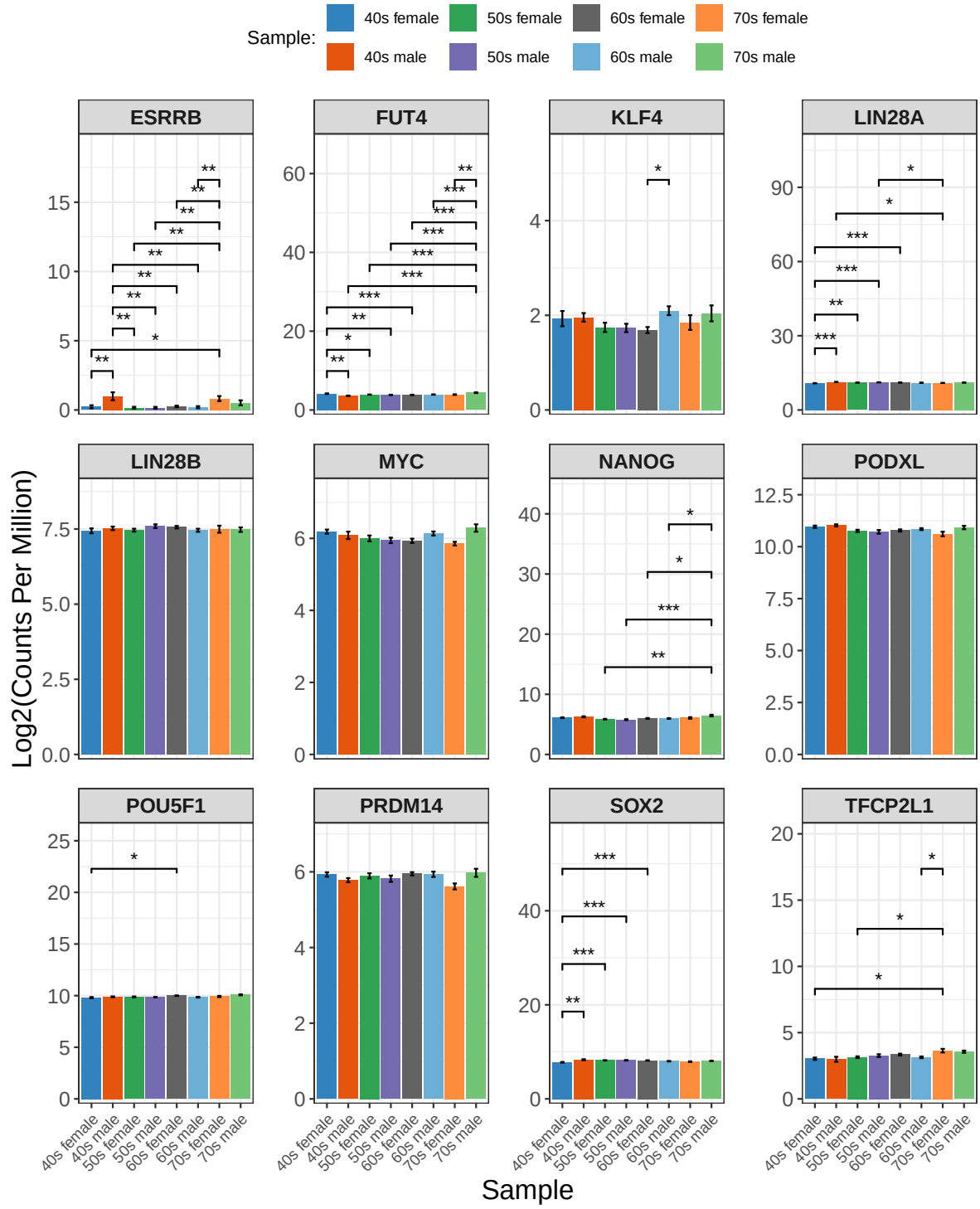


Figure 6: Expression of Pluripotency Genes

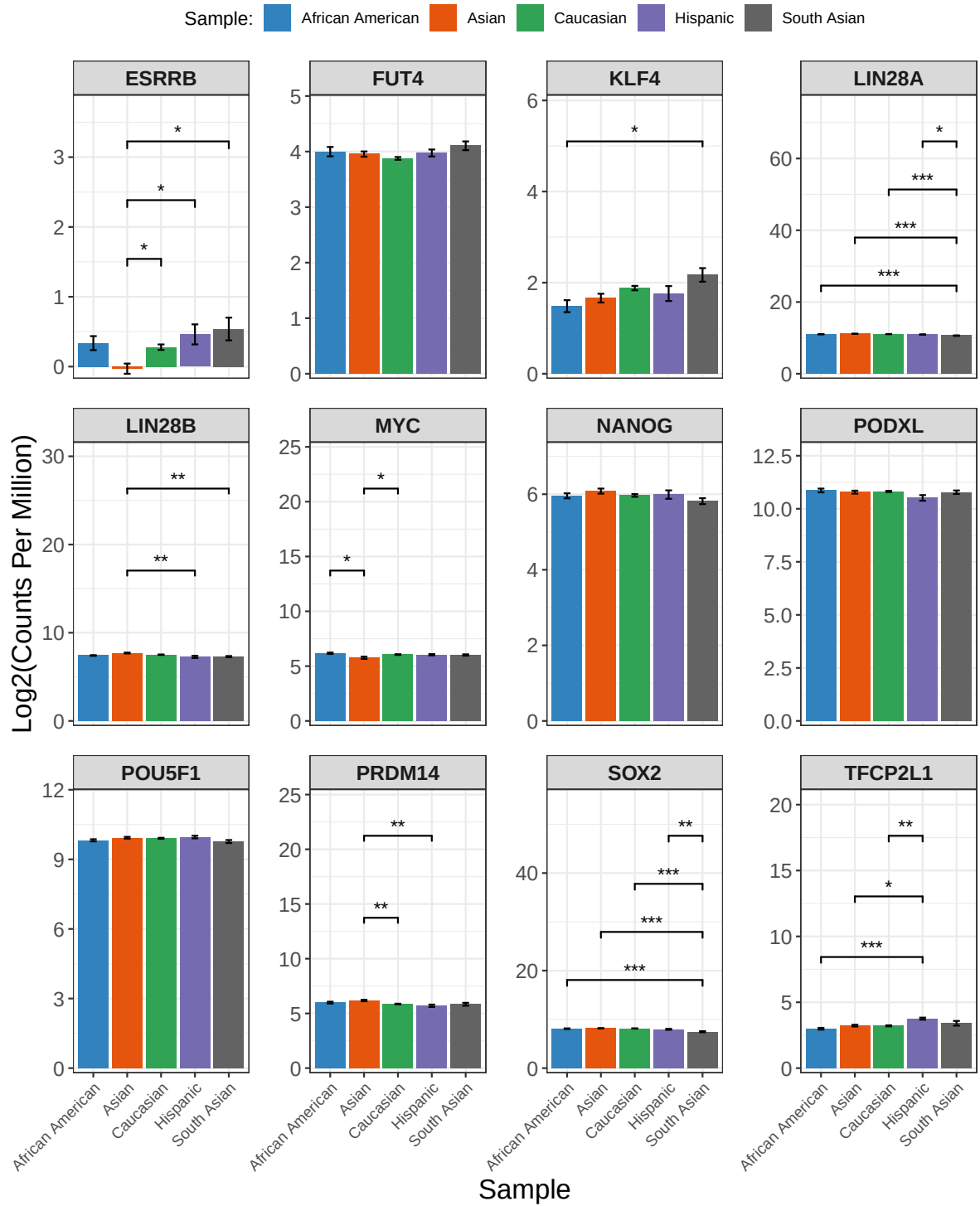


Figure 7: Expression of Pluripotency Genes

age_sex50s female	-0.02437	0.02884	3485.00000	-0.845	0.3983
age_sex50s male	-0.02283	0.03182	3485.00000	-0.718	0.4731
age_sex60s female	0.01962	0.02818	3485.00000	0.696	0.4863
age_sex60s male	0.01738	0.02962	3485.00000	0.587	0.5574
age_sex70s female	0.02017	0.04233	3485.00000	0.476	0.6339
age_sex70s male	0.20776	0.04614	3485.00000	4.503	6.92e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	ag_40m	ag_50f	ag_50m	ag_60f	ag_60m	ag_70f
age_sx40sml	-0.011						
ag_sx50sfml	-0.019	0.371					
age_sx50sml	-0.018	0.336	0.608				
ag_sx60sfml	-0.020	0.379	0.686	0.622			
age_sx60sml	-0.019	0.361	0.653	0.591	0.668		
ag_sx70sfml	-0.013	0.253	0.457	0.414	0.467	0.445	
age_sx70sml	-0.012	0.232	0.419	0.380	0.429	0.408	0.285

Type III Analysis of Variance Table with Satterthwaite's method

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
age_sex	7.7171	1.1024	7	3485	5.3154	4.615e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

\$emmeans

age_sex	emmean	SE	df	asympt.LCL	asympt.UCL
40s female	6.20	1	Inf	4.23	8.16
40s male	6.31	1	Inf	4.35	8.27
50s female	6.17	1	Inf	4.21	8.13
50s male	6.17	1	Inf	4.21	8.13
60s female	6.21	1	Inf	4.25	8.18
60s male	6.21	1	Inf	4.25	8.17
70s female	6.22	1	Inf	4.25	8.18
70s male	6.40	1	Inf	4.44	8.36

Degrees-of-freedom method: asymptotic

Confidence level used: 0.95

\$contrasts

contrast	estimate	SE	df	z.ratio	p.value
40s female - 40s male	-0.11258	0.0521	Inf	-2.159	0.3766

40s female - 50s female	0.02437	0.0288	Inf	0.845	0.9905
40s female - 50s male	0.02283	0.0318	Inf	0.718	0.9965
40s female - 60s female	-0.01962	0.0282	Inf	-0.696	0.9971
40s female - 60s male	-0.01738	0.0296	Inf	-0.587	0.9990
40s female - 70s female	-0.02017	0.0423	Inf	-0.476	0.9998
40s female - 70s male	-0.20776	0.0461	Inf	-4.503	0.0002
40s male - 50s female	0.13695	0.0493	Inf	2.775	0.1012
40s male - 50s male	0.13541	0.0511	Inf	2.648	0.1390
40s male - 60s female	0.09296	0.0490	Inf	1.899	0.5518
40s male - 60s male	0.09520	0.0498	Inf	1.911	0.5429
40s male - 70s female	0.09242	0.0583	Inf	1.586	0.7589
40s male - 70s male	-0.09518	0.0611	Inf	-1.558	0.7754
50s female - 50s male	-0.00153	0.0270	Inf	-0.057	1.0000
50s female - 60s female	-0.04399	0.0226	Inf	-1.946	0.5190
50s female - 60s male	-0.04175	0.0244	Inf	-1.712	0.6792
50s female - 70s female	-0.04453	0.0388	Inf	-1.146	0.9465
50s female - 70s male	-0.23213	0.0430	Inf	-5.403	<.0001
50s male - 60s female	-0.04246	0.0263	Inf	-1.614	0.7418
50s male - 60s male	-0.04021	0.0278	Inf	-1.445	0.8364
50s male - 70s female	-0.04300	0.0411	Inf	-1.046	0.9673
50s male - 70s male	-0.23059	0.0450	Inf	-5.123	<.0001
60s female - 60s male	0.00224	0.0236	Inf	0.095	1.0000
60s female - 70s female	-0.00054	0.0384	Inf	-0.014	1.0000
60s female - 70s male	-0.18814	0.0425	Inf	-4.425	0.0003
60s male - 70s female	-0.00278	0.0394	Inf	-0.071	1.0000
60s male - 70s male	-0.19038	0.0435	Inf	-4.378	0.0003
70s female - 70s male	-0.18760	0.0530	Inf	-3.542	0.0095

Degrees-of-freedom method: asymptotic

P value adjustment: tukey method for comparing a family of 8 estimates

A tibble: 24 x 8

	term	df	sumsq	meansq	statistic	p.value	Gene	p_adj
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>
1	group_var	7	3.16	0.452	6.07	0.00000129	LIN28A	0.00000517
2	Residuals	284	21.1	0.0744	NA	NA	LIN28A	NA
3	group_var	7	7.06	1.01	3.48	0.00136	TFCP2L1	0.00272
4	Residuals	284	82.3	0.290	NA	NA	TFCP2L1	NA
5	group_var	7	5.84	0.835	6.47	0.000000437	SOX2	0.00000262
6	Residuals	284	36.6	0.129	NA	NA	SOX2	NA
7	group_var	7	1.63	0.232	3.04	0.00420	POU5F1	0.00720
8	Residuals	284	21.7	0.0763	NA	NA	POU5F1	NA

```

  9 group_var      7  0.898 0.128      0.888 0.516      LIN28B 0.516
10 Residuals    284 41.0  0.144      NA    NA      LIN28B NA
# i 14 more rows

# A tibble: 336 x 7
# Groups:   Gene [12]
      diff      lwr      upr      p_value Gene comparison      p_adj
    <dbl>    <dbl>    <dbl>    <dbl> <chr>    <chr>      <dbl>
1  0.467    0.137    0.798  0.000565 LIN28A 40s male-40s female 0.00528
2  0.238    0.0551   0.421  0.00225  LIN28A 50s female-40s female 0.0157
3  0.353    0.152    0.555  0.00000497 LIN28A 50s male-40s female 0.000139
4  0.266    0.0874   0.444  0.000214  LIN28A 60s female-40s female 0.00299
5  0.179   -0.00848  0.367  0.0732   LIN28A 60s male-40s female 0.256
6  0.0928  -0.175    0.361  0.965    LIN28A 70s female-40s female 1.000
7  0.249   -0.0428  0.542  0.158    LIN28A 70s male-40s female 0.442
8 -0.229   -0.542    0.0832  0.331    LIN28A 50s female-40s male 0.842
9 -0.114   -0.438    0.210  0.962    LIN28A 50s male-40s male 1.000
10 -0.201  -0.512    0.109  0.496    LIN28A 60s female-40s male 0.992
# i 326 more rows

```

Residual variance = 0.2103 and intercept variance = 12.0276 means most of the variation is between genes, not sex and age.

Fixed effects Sex effect (male): Expression is slightly higher in males, for all genes, by 0.036 logCPM; significant at $p = 0.0205$, but only at 0.03648 units (log2 count per million/relative expression?).

Age effects: Only the 70s group is significantly higher than 40s ($p = 0.0135$), 0.08018 units. 50s and 60s are not significantly different from 40s.

Type III Analysis of Variance Table with Satterthwaite's method - confirms both sex and age_decade have a significant effect on pluripotency gene expression when averaged across all genes.

Predicted mean logCPM expression (adjusted means, not raw means)

female: 6.20. male: 6.23

40s: 6.20, 50s: 6.18, 60s: 6.21, 70s: 6.28

Overall, pluripotency gene expression is slightly higher in males compared to females, regardless of age. Expression also shows an increasing trend with age, with samples from individuals in their 70s being significantly higher than those from individuals in their 40s. Differences between other age decades (50s, 60s) and 40s were not statistically significant.

```
#pluripotency genes, race stats

df_pluri_race <- pluri_race_results$data
# Fit mixed model
#model the response variable, Expression, for the fixed effects, age and sex
# genes are treated as a random effect, all with different y intercepts,
# since expression levels vary by gene
lmm_pluri_race <- lmer(Expression ~ race + (1 | `Gene name`), data = df_pluri_race)
summary_pluri_race <- summary(lmm_pluri_race)
# type III ANOVA to test significance of fixed effects on expression
lmm_pluri_race_anova <- anova(lmm_pluri_race, type = 3)

# estimated marginal means (or least-squares means) with pairwise comparisons
lmm_pluri_race_emm <- emmeans(lmm_pluri_race, pairwise ~ race)
```

Note: D.f. calculations have been disabled because the number of observations exceeds 3000. To enable adjustments, add the argument 'pbkrtest.limit = 3504' (or larger) [or, globally, 'set emm_options(pbkrtest.limit = 3504)' or larger]; but be warned that this may result in large computation time and memory use.

Note: D.f. calculations have been disabled because the number of observations exceeds 3000. To enable adjustments, add the argument 'lmerTest.limit = 3504' (or larger) [or, globally, 'set emm_options(lmerTest.limit = 3504)' or larger]; but be warned that this may result in large computation time and memory use.

```
#lmm summary
summary_pluri_race
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: Expression ~ race + (1 | `Gene name`)
Data: df_pluri_race
```

REML criterion at convergence: 4592.4

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-4.7200	-0.5553	0.0448	0.5467	5.5487

Random effects:

Groups	Name	Variance	Std.Dev.
--------	------	----------	----------


```
Gene name (Intercept) 12.0051 3.4648
Residual              0.2092 0.4574
Number of obs: 3504, groups: Gene name, 12
```

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	6.17921	1.00065	11.01802	6.175	6.91e-05 ***
raceAsian	0.03317	0.03742	3488.00000	0.886	0.375
raceCaucasian	0.03371	0.03089	3488.00000	1.092	0.275
raceHispanic	0.01311	0.04601	3488.00000	0.285	0.776
raceSouth Asian	-0.02661	0.04704	3488.00000	-0.566	0.572

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	racAsn	rcCcsn	rcHspn
raceAsian	-0.023			
raceCaucasn	-0.028	0.754		
raceHispanic	-0.019	0.506	0.613	
raceSothAsn	-0.019	0.495	0.600	0.403

```
#anova
lmm_pluri_race_anova
```

Type III Analysis of Variance Table with Satterthwaite's method

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
race	0.7839	0.19598	4	3488	0.9367	0.4415

```
#means
lmm_pluri_race_emm
```

\$emmeans

race	emmean	SE	df	asympt.LCL	asympt.UCL
African American	6.18	1	Inf	4.22	8.14
Asian	6.21	1	Inf	4.25	8.17
Caucasian	6.21	1	Inf	4.25	8.17
Hispanic	6.19	1	Inf	4.23	8.15
South Asian	6.15	1	Inf	4.19	8.11

Degrees-of-freedom method: asymptotic

Confidence level used: 0.95

```
$contrasts
contrast      estimate      SE  df z.ratio p.value
African American - Asian    -0.03317 0.0374 Inf  -0.886  0.9021
African American - Caucasian -0.03371 0.0309 Inf  -1.092  0.8110
African American - Hispanic -0.01311 0.0460 Inf  -0.285  0.9986
African American - South Asian 0.02661 0.0470 Inf   0.566  0.9800
Asian - Caucasian    -0.00055 0.0247 Inf  -0.022  1.0000
Asian - Hispanic      0.02006 0.0421 Inf   0.476  0.9895
Asian - South Asian    0.05977 0.0432 Inf   1.382  0.6390
Caucasian - Hispanic    0.02061 0.0364 Inf   0.566  0.9800
Caucasian - South Asian 0.06032 0.0377 Inf   1.599  0.4980
Hispanic - South Asian  0.03972 0.0509 Inf   0.781  0.9362
```

Degrees-of-freedom method: asymptotic

P value adjustment: tukey method for comparing a family of 5 estimates

```
#gene anova FDR corrected
pluri_race_results$anova_fdr
```

```
# A tibble: 24 x 8
  term      df  sumsq meansq statistic      p.value Gene      p_adj
  <chr>    <dbl> <dbl>  <dbl>    <dbl>    <dbl> <chr>    <dbl>
1 group_var      4  2.66  0.665      8.83 0.0000000983 LIN28A 0.00000590
2 Residuals    287 21.6  0.0754      NA    NA      LIN28A NA
3 group_var      4  5.49  1.37      4.70 0.00110      TFCEP2L1 0.00270
4 Residuals    287 83.9  0.292      NA    NA      TFCEP2L1 NA
5 group_var      4  5.83  1.46     11.4 0.0000000131 SOX2    0.000000157
6 Residuals    287 36.7  0.128      NA    NA      SOX2    NA
7 group_var      4  0.444 0.111      1.39 0.236      POU5F1  0.258
8 Residuals    287 22.9  0.0796      NA    NA      POU5F1  NA
9 group_var      4  2.76  0.691      5.06 0.000589     LIN28B  0.00236
10 Residuals    287 39.2  0.136      NA    NA     LIN28B  NA
# i 14 more rows
```

```
#gene tueky corrected
pluri_race_results$tukey_fdr
```

```
# A tibble: 120 x 7
# Groups:   Gene [12]
  diff      lwr      upr      p_value Gene      comparison      p_adj
```

	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<chr>	<dbl>
1	0.112	-0.101	0.326	0.602	LIN28A	Asian-African American	7.52e-1
2	0.0313	-0.145	0.208	0.989	LIN28A	Caucasian-African American	9.89e-1
3	-0.0898	-0.352	0.173	0.882	LIN28A	Hispanic-African American	9.80e-1
4	-0.390	-0.659	-0.122	0.000791	LIN28A	South Asian-African Americ~	2.64e-3
5	-0.0808	-0.222	0.0602	0.516	LIN28A	Caucasian-Asian	7.38e-1
6	-0.202	-0.442	0.0385	0.146	LIN28A	Hispanic-Asian	2.92e-1
7	-0.502	-0.749	-0.256	0.000000528	LIN28A	South Asian-Asian	5.28e-6
8	-0.121	-0.329	0.0869	0.500	LIN28A	Hispanic-Caucasian	7.38e-1
9	-0.422	-0.637	-0.206	0.00000157	LIN28A	South Asian-Caucasian	7.86e-6
10	-0.301	-0.591	-0.0103	0.0383	LIN28A	South Asian-Hispanic	9.57e-2

i 110 more rows

Differentiation Markers

Expression of Differentiation Genes (ectoderm: PAX6, NESTIN (NES) or SOX1 (neural progenitor/stem cell markers), KRT14 (epithelial), endoderm: SOX17, AFP (hepatic), PDX1 (pancreatic), mesoderm: CD34 (hematopoietic), CD90 (THY1)(mesenchymal), NKX2.5 (cardiac progenitor)).

need to fix significance bars + asterisks when values are negative

Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]
 Formula: Expression ~ age_sex + (1 | `Gene name`)
 Data: df_diff_agesex

REML criterion at convergence: 8696.2

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.6292	-0.4208	-0.1085	0.2793	7.1639

Random effects:

Groups	Name	Variance	Std.Dev.
Gene name	(Intercept)	12.362	3.516
Residual		1.114	1.055

Number of obs: 2920, groups: Gene name, 10

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	2.16618	1.11348	9.04685	1.945	0.0834 .

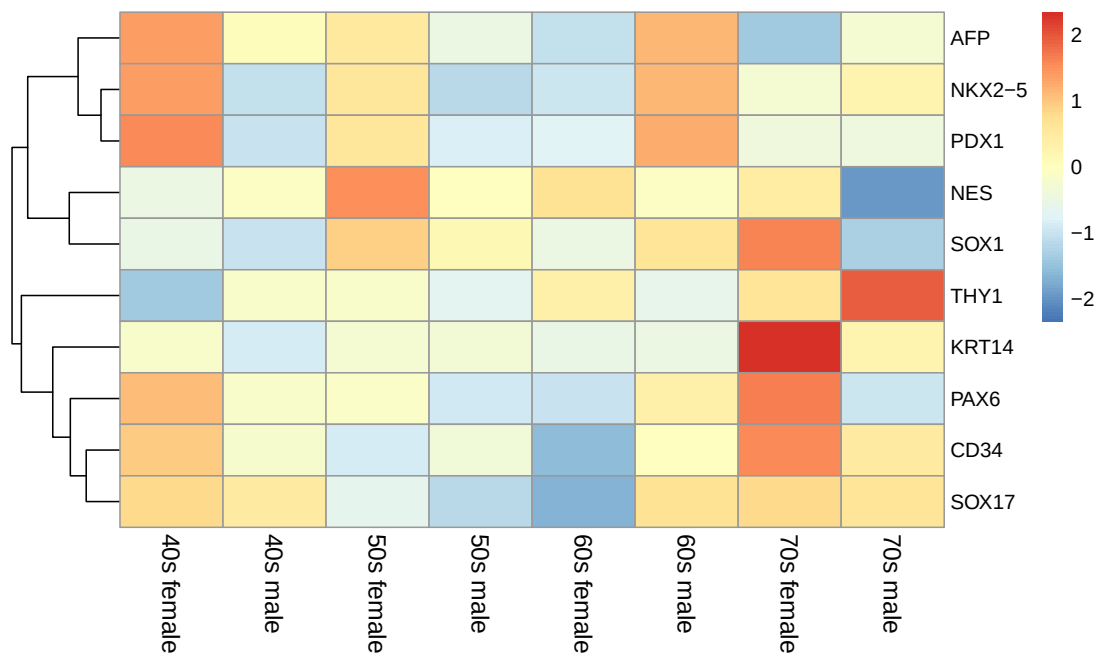


Figure 8: Expression of Differentiation Genes

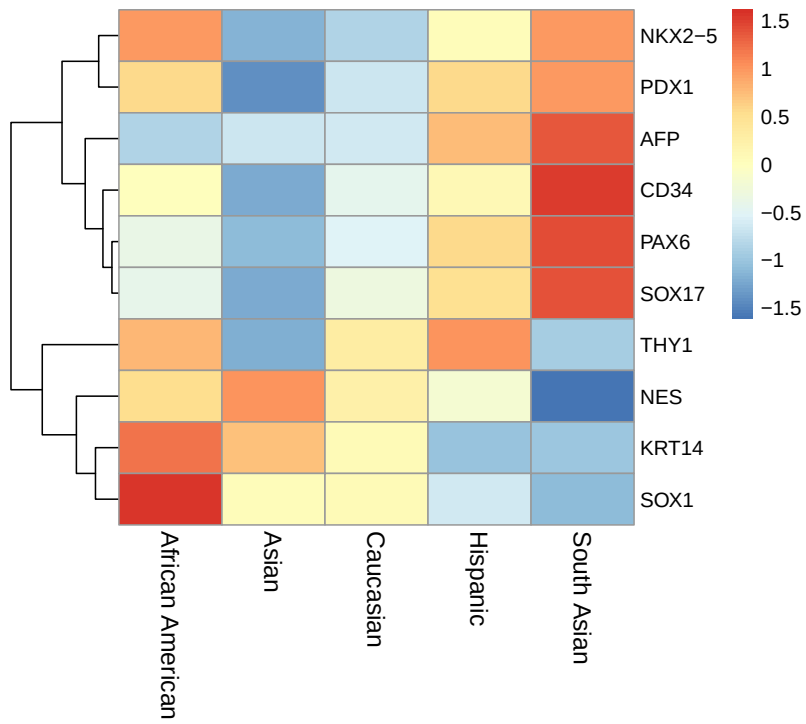


Figure 9: Expression of Differentiation Genes

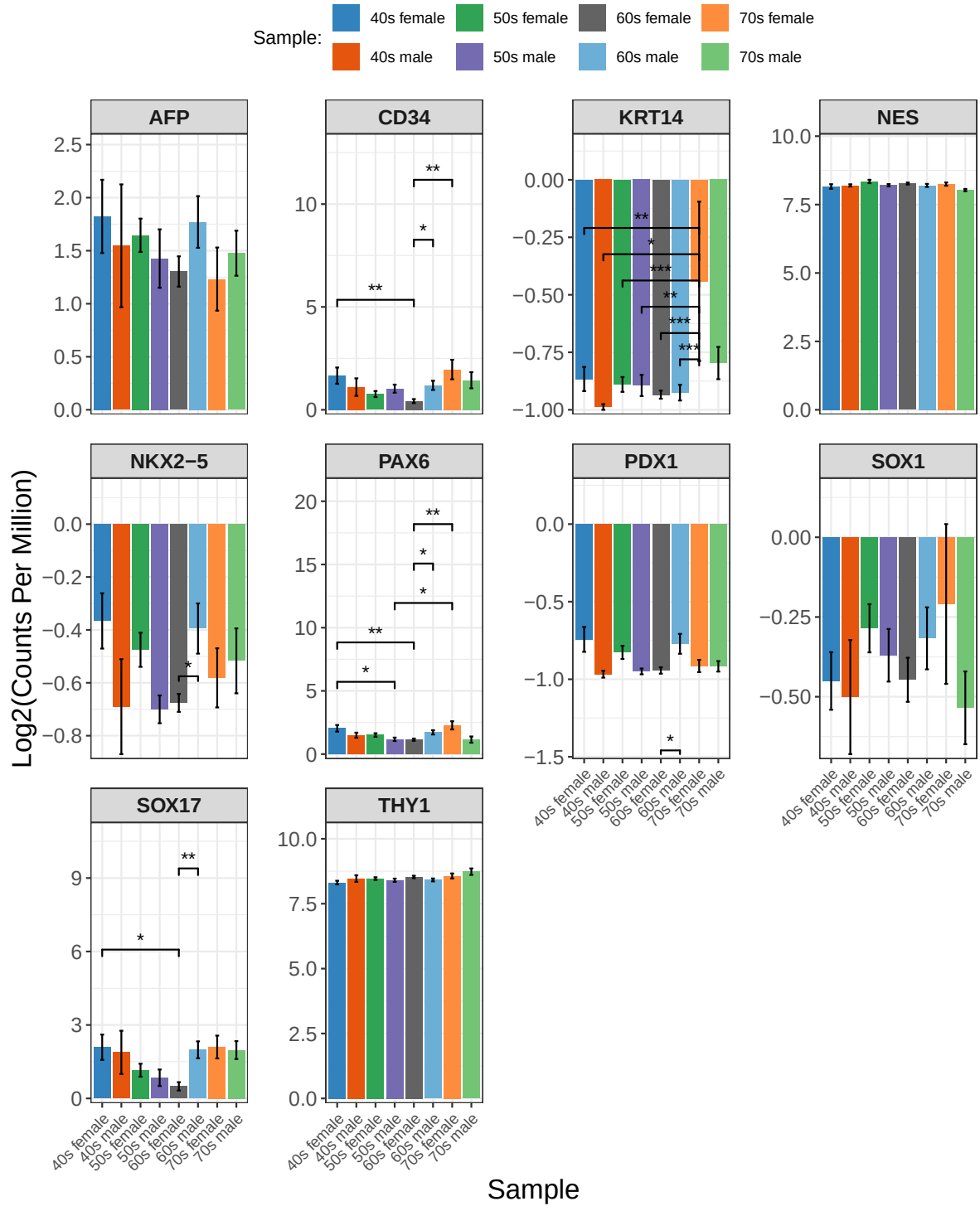


Figure 10: Expression of Differentiation Genes

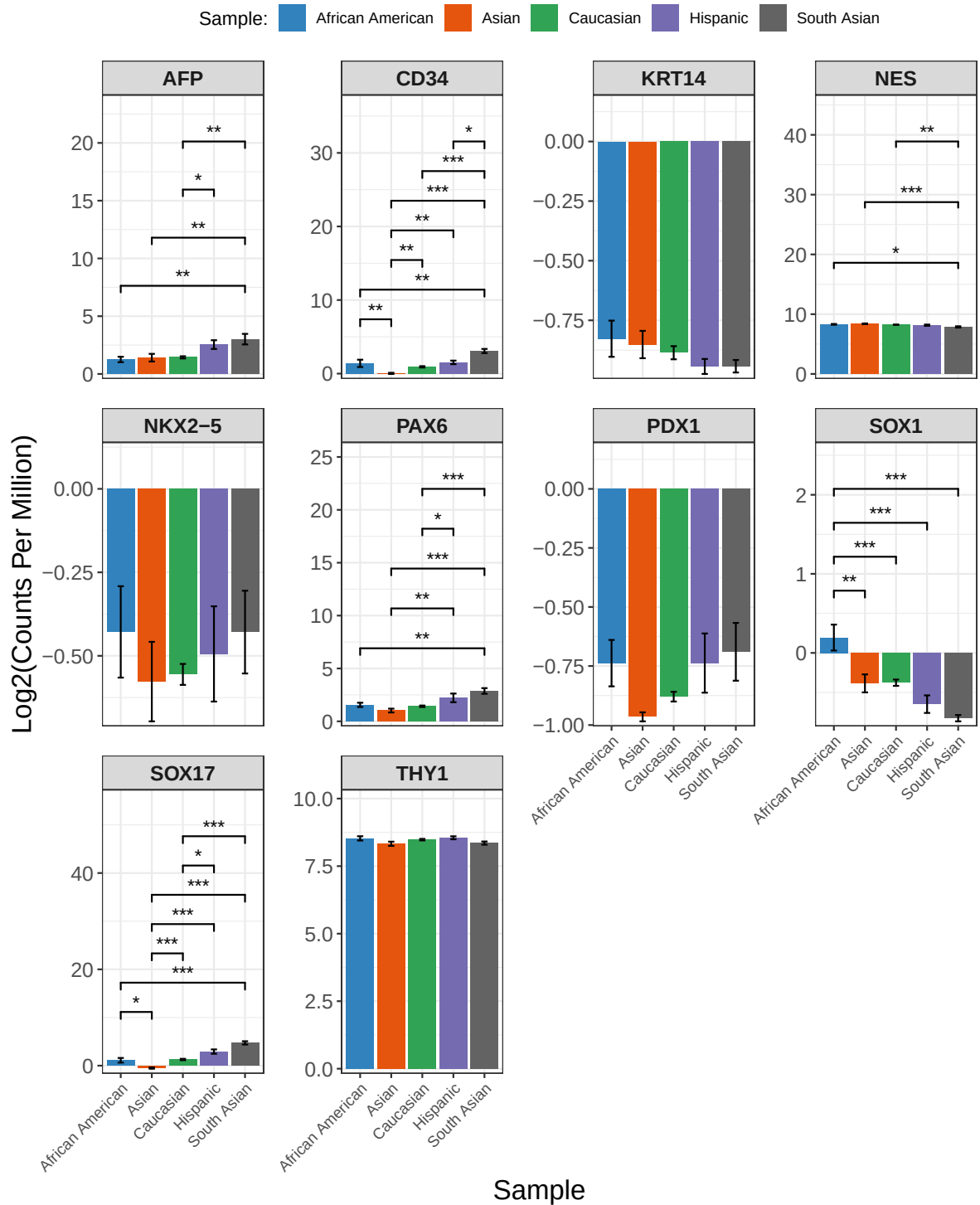


Figure 11: Expression of Differentiation Genes

age_sex40s male	-0.21109	0.13236	2903.00000	-1.595	0.1109	
age_sex50s female	-0.22545	0.07323	2903.00000	-3.079	0.0021	**
age_sex50s male	-0.34974	0.08078	2903.00000	-4.329	1.55e-05	***
age_sex60s female	-0.44996	0.07155	2903.00000	-6.288	3.69e-10	***
age_sex60s male	-0.07894	0.07521	2903.00000	-1.050	0.2940	
age_sex70s female	0.05717	0.10748	2903.00000	0.532	0.5948	
age_sex70s male	-0.16292	0.11714	2903.00000	-1.391	0.1644	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	ag_40m	ag_50f	ag_50m	ag_60f	ag_60m	ag_70f
age_sx40sml	-0.024						
ag_sx50sfml	-0.044	0.371					
age_sx50sml	-0.040	0.336	0.608				
ag_sx60sfml	-0.045	0.379	0.686	0.622			
age_sx60sml	-0.043	0.361	0.653	0.591	0.668		
ag_sx70sfml	-0.030	0.253	0.457	0.414	0.467	0.445	
age_sx70sml	-0.028	0.232	0.419	0.380	0.429	0.408	0.285

Type III Analysis of Variance Table with Satterthwaite's method

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
age_sex	81.569	11.653	7	2903	10.46	4.893e-13 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

\$emmeans

age_sex	emmean	SE	df	lower.CL	upper.CL
40s female	2.17	1.11	9.05	-0.351	4.68
40s male	1.96	1.12	9.20	-0.566	4.48
50s female	1.94	1.11	9.02	-0.575	4.46
50s male	1.82	1.11	9.04	-0.700	4.33
60s female	1.72	1.11	9.02	-0.800	4.23
60s male	2.09	1.11	9.02	-0.429	4.60
70s female	2.22	1.12	9.11	-0.295	4.74
70s male	2.00	1.12	9.14	-0.516	4.52

Degrees-of-freedom method: kenward-roger

Confidence level used: 0.95

\$contrasts

contrast	estimate	SE	df	t.ratio	p.value
----------	----------	----	----	---------	---------

40s female - 40s male	0.2111	0.1320	2903	1.595	0.7536
40s female - 50s female	0.2255	0.0732	2903	3.079	0.0437
40s female - 50s male	0.3497	0.0808	2903	4.329	0.0004
40s female - 60s female	0.4500	0.0716	2903	6.288	<.0001
40s female - 60s male	0.0789	0.0752	2903	1.050	0.9667
40s female - 70s female	-0.0572	0.1070	2903	-0.532	0.9995
40s female - 70s male	0.1629	0.1170	2903	1.391	0.8617
40s male - 50s female	0.0144	0.1250	2903	0.115	1.0000
40s male - 50s male	0.1386	0.1300	2903	1.068	0.9634
40s male - 60s female	0.2389	0.1240	2903	1.922	0.5360
40s male - 60s male	-0.1322	0.1260	2903	-1.045	0.9674
40s male - 70s female	-0.2683	0.1480	2903	-1.813	0.6109
40s male - 70s male	-0.0482	0.1550	2903	-0.311	1.0000
50s female - 50s male	0.1243	0.0686	2903	1.813	0.6113
50s female - 60s female	0.2245	0.0574	2903	3.911	0.0024
50s female - 60s male	-0.1465	0.0619	2903	-2.367	0.2581
50s female - 70s female	-0.2826	0.0986	2903	-2.866	0.0800
50s female - 70s male	-0.0625	0.1090	2903	-0.573	0.9992
50s male - 60s female	0.1002	0.0668	2903	1.501	0.8072
50s male - 60s male	-0.2708	0.0707	2903	-3.832	0.0033
50s male - 70s female	-0.4069	0.1040	2903	-3.899	0.0025
50s male - 70s male	-0.1868	0.1140	2903	-1.635	0.7291
60s female - 60s male	-0.3710	0.0599	2903	-6.193	<.0001
60s female - 70s female	-0.5071	0.0974	2903	-5.208	<.0001
60s female - 70s male	-0.2870	0.1080	2903	-2.659	0.1358
60s male - 70s female	-0.1361	0.1000	2903	-1.360	0.8753
60s male - 70s male	0.0840	0.1100	2903	0.761	0.9950
70s female - 70s male	0.2201	0.1340	2903	1.637	0.7280

Degrees-of-freedom method: kenward-roger

P value adjustment: tukey method for comparing a family of 8 estimates

A tibble: 20 x 8

	term	df	sumsq	meansq	statistic	p.value	Gene	p_adj
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>
1	group_var	7	1.57	0.224	1.56	0.146	NES	0.183
2	Residuals	284	40.7	0.143	NA	NA	NES	NA
3	group_var	7	58.1	8.30	4.34	0.000139	CD34	0.000694
4	Residuals	284	543.	1.91	NA	NA	CD34	NA
5	group_var	7	12.0	1.71	0.772	0.612	AFP	0.612
6	Residuals	284	631.	2.22	NA	NA	AFP	NA
7	group_var	7	4.88	0.697	2.92	0.00581	NKX2-5	0.0116

8	Residuals	284	67.8	0.239	NA	NA	NKX2-5	NA
9	group_var	7	118.	16.9	3.78	0.000617	SOX17	0.00206
10	Residuals	284	1268.	4.46	NA	NA	SOX17	NA
11	group_var	7	34.9	4.98	4.56	0.0000771	PAX6	0.000694
12	Residuals	284	310.	1.09	NA	NA	PAX6	NA
13	group_var	7	2.23	0.319	2.01	0.0544	THY1	0.0778
14	Residuals	284	45.1	0.159	NA	NA	THY1	NA
15	group_var	7	1.89	0.270	2.76	0.00865	PDX1	0.0144
16	Residuals	284	27.8	0.0977	NA	NA	PDX1	NA
17	group_var	7	2.03	0.291	0.777	0.607	SOX1	0.612
18	Residuals	284	106.	0.374	NA	NA	SOX1	NA
19	group_var	7	3.20	0.458	3.52	0.00121	KRT14	0.00302
20	Residuals	284	36.9	0.130	NA	NA	KRT14	NA

A tibble: 280 x 7

Groups: Gene [10]

	diff	lwr	upr	p_value	Gene	comparison	p_adj
	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<chr>	<dbl>
1	0.0402	-0.418	0.499	1.000	NES	40s male-40s female	1
2	0.183	-0.0704	0.437	0.351	NES	50s female-40s female	1
3	0.0455	-0.234	0.325	1.000	NES	50s male-40s female	1
4	0.109	-0.139	0.357	0.884	NES	60s female-40s female	1
5	0.0395	-0.221	0.300	1.000	NES	60s male-40s female	1
6	0.0876	-0.285	0.460	0.996	NES	70s female-40s female	1
7	-0.131	-0.537	0.275	0.976	NES	70s male-40s female	1
8	0.143	-0.291	0.577	0.973	NES	50s female-40s male	1
9	0.00526	-0.445	0.455	1.000	NES	50s male-40s male	1
10	0.0684	-0.362	0.499	1.000	NES	60s female-40s male	1

i 270 more rows

#diff genes, race stats

```
df_diff_race <- diff_race_results$data
```

```
# Fit mixed model
```

```
#model the response variable, Expression, for the fixed effects, age and sex
```

```
# genes are treated as a random effect, all with different y intercepts,
```

```
# since expression levels vary by gene
```

```
lmm_diff_race <- lmer(Expression ~ race + (1 | `Gene name`), data = df_diff_race)
```

```
summary_diff_race <- summary(lmm_diff_race)
```

```
# type III ANOVA to test significance of fixed effects on expression
```

```
lmm_diff_race_anova <- anova(lmm_diff_race, type = 3)
```

```
# estimated marginal means (or least-squares means) with pairwise comparisons
lmm_diff_race_emm <- emmeans(lmm_diff_race, pairwise ~ race)

#lmm summary
summary_diff_race
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: Expression ~ race + (1 | `Gene name`)
Data: df_diff_race
```

REML criterion at convergence: 8633.6

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.4076	-0.4522	-0.0864	0.2784	7.4030

Random effects:

Groups	Name	Variance	Std.Dev.
Gene name	(Intercept)	12.363	3.516
	Residual	1.093	1.046

Number of obs: 2920, groups: Gene name, 10

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	2.03509	1.11433	9.07427	1.826	0.1008
raceAsian	-0.44469	0.09369	2906.00000	-4.746	2.17e-06 ***
raceCaucasian	-0.12665	0.07734	2906.00000	-1.638	0.1016
raceHispanic	0.27489	0.11521	2906.00000	2.386	0.0171 *
raceSouth Asian	0.66696	0.11779	2906.00000	5.662	1.64e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	racAsn	rcCcsn	rcHspn
raceAsian	-0.052			
raceCaucasn	-0.063	0.754		
raceHispanc	-0.043	0.506	0.613	
raceSothAsn	-0.042	0.495	0.600	0.403

```
#anova
lmm_diff_race_anova
```

```
Type III Analysis of Variance Table with Satterthwaite's method
      Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
race 138.86  34.715     4  2906  31.756 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#means
lmm_diff_race_emm
```

```
$emmeans
      race      emmean    SE    df lower.CL upper.CL
African American    2.04 1.11 9.07   -0.483    4.55
Asian                1.59 1.11 9.04   -0.926    4.11
Caucasian            1.91 1.11 9.00   -0.607    4.42
Hispanic             2.31 1.12 9.11   -0.209    4.83
South Asian          2.70 1.12 9.12    0.183    5.22
```

```
Degrees-of-freedom method: kenward-roger
Confidence level used: 0.95
```

```
$contrasts
      contrast      estimate    SE    df t.ratio p.value
African American - Asian      0.445 0.0937 2906   4.746 <.0001
African American - Caucasian  0.127 0.0773 2906   1.638 0.4733
African American - Hispanic  -0.275 0.1150 2906  -2.386 0.1194
African American - South Asian -0.667 0.1180 2906  -5.662 <.0001
Asian - Caucasian            -0.318 0.0619 2906  -5.140 <.0001
Asian - Hispanic             -0.720 0.1050 2906  -6.824 <.0001
Asian - South Asian          -1.112 0.1080 2906 -10.268 <.0001
Caucasian - Hispanic         -0.402 0.0912 2906  -4.401 0.0001
Caucasian - South Asian      -0.794 0.0945 2906  -8.401 <.0001
Hispanic - South Asian       -0.392 0.1270 2906  -3.079 0.0179
```

```
Degrees-of-freedom method: kenward-roger
P value adjustment: tukey method for comparing a family of 5 estimates
```

```
#gene anova FDR corrected
diff_race_results$anova_fdr
```

```
# A tibble: 20 x 8
```

	term	df	sumsq	meansq	statistic	p.value	Gene	p_adj
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>
1	group_var	4	2.74	0.685	4.97	6.95e- 4	NES	1.16e- 3
2	Residuals	287	39.6	0.138	NA	NA	NES	NA
3	group_var	4	95.0	23.7	13.5	4.57e-10	CD34	2.28e- 9
4	Residuals	287	506.	1.76	NA	NA	CD34	NA
5	group_var	4	46.7	11.7	5.63	2.26e- 4	AFP	4.52e- 4
6	Residuals	287	596.	2.08	NA	NA	AFP	NA
7	group_var	4	0.536	0.134	0.533	7.12e- 1	NKX2-5	7.91e- 1
8	Residuals	287	72.2	0.251	NA	NA	NKX2-5	NA
9	group_var	4	297.	74.4	19.6	2.82e-14	SOX17	2.82e-13
10	Residuals	287	1089.	3.79	NA	NA	SOX17	NA
11	group_var	4	40.6	10.2	9.56	2.85e- 7	PAX6	9.50e- 7
12	Residuals	287	305.	1.06	NA	NA	PAX6	NA
13	group_var	4	1.01	0.252	1.56	1.85e- 1	THY1	2.31e- 1
14	Residuals	287	46.3	0.161	NA	NA	THY1	NA
15	group_var	4	1.33	0.331	3.36	1.04e- 2	PDX1	1.49e- 2
16	Residuals	287	28.3	0.0987	NA	NA	PDX1	NA
17	group_var	4	10.2	2.54	7.43	1.05e- 5	SOX1	2.61e- 5
18	Residuals	287	98.1	0.342	NA	NA	SOX1	NA
19	group_var	4	0.194	0.0485	0.348	8.45e- 1	KRT14	8.45e- 1
20	Residuals	287	39.9	0.139	NA	NA	KRT14	NA

```
#gene tueky corrected
diff_race_results$tukey_fdr
```

```
# A tibble: 100 x 7
```

```
# Groups: Gene [10]
```

	diff	lwr	upr	p_value	Gene	comparison	p_adj
	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<chr>	<dbl>
1	0.0937	-0.195	0.383	0.900	NES	Asian-African American	0.964
2	-0.0578	-0.296	0.181	0.964	NES	Caucasian-African American	0.964
3	-0.143	-0.499	0.212	0.803	NES	Hispanic-African American	0.964
4	-0.431	-0.794	-0.0680	0.0109	NES	South Asian-African American	0.0363
5	-0.152	-0.342	0.0393	0.190	NES	Caucasian-Asian	0.447
6	-0.237	-0.562	0.0881	0.268	NES	Hispanic-Asian	0.447
7	-0.525	-0.859	-0.191	0.000211	NES	South Asian-Asian	0.00211

```

8 -0.0855 -0.367 0.196 0.920 NES Hispanic-Caucasian 0.964
9 -0.373 -0.665 -0.0821 0.00455 NES South Asian-Caucasian 0.0227
10 -0.288 -0.681 0.105 0.262 NES South Asian-Hispanic 0.447
# i 90 more rows

```

Housekeeping Genes

Figure 5. Expression of Housekeeping Genes (some sort of general figure or sorted by house-keeping type (i.e. ribosomal, metabolic, cytoskeleton, etc.))

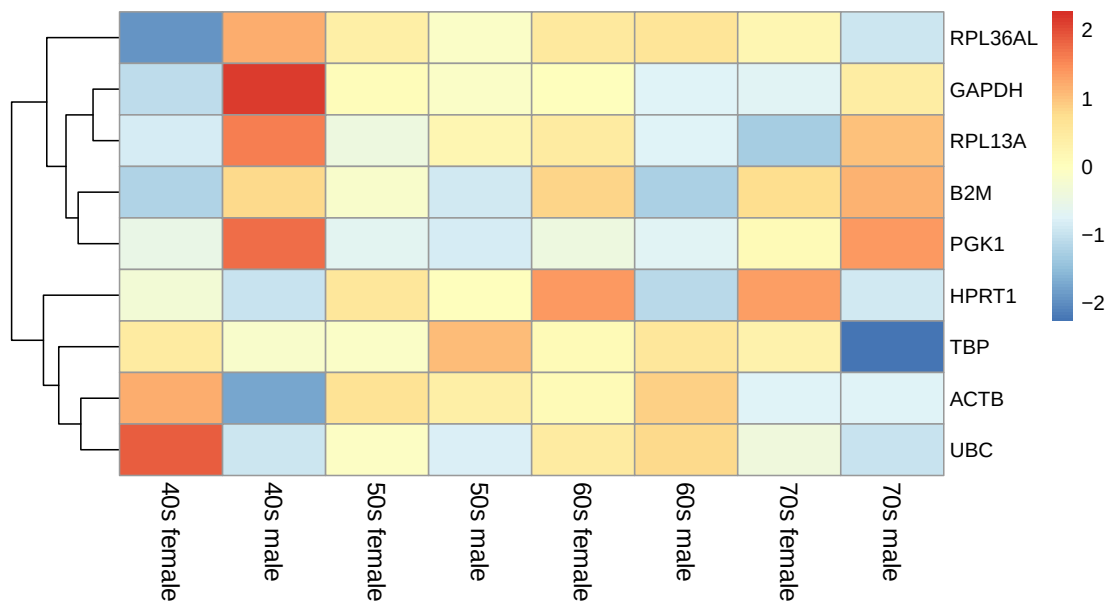


Figure 12: Expression of Housekeeping Genes

```

# A tibble: 18 x 8
  term      df  sumsq meansq statistic  p.value Gene      p_adj
  <chr>    <dbl> <dbl> <dbl>    <dbl>   <dbl> <chr>    <dbl>
1 group_var      7  0.496 0.0709      3.06 4.04e-3 TBP      0.0121

```

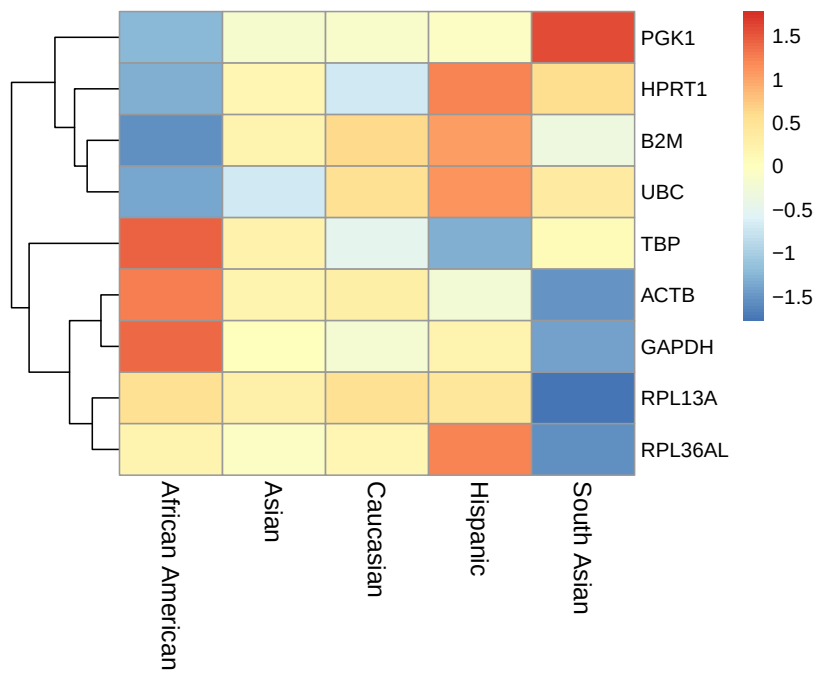


Figure 13: Expression of Housekeeping Genes

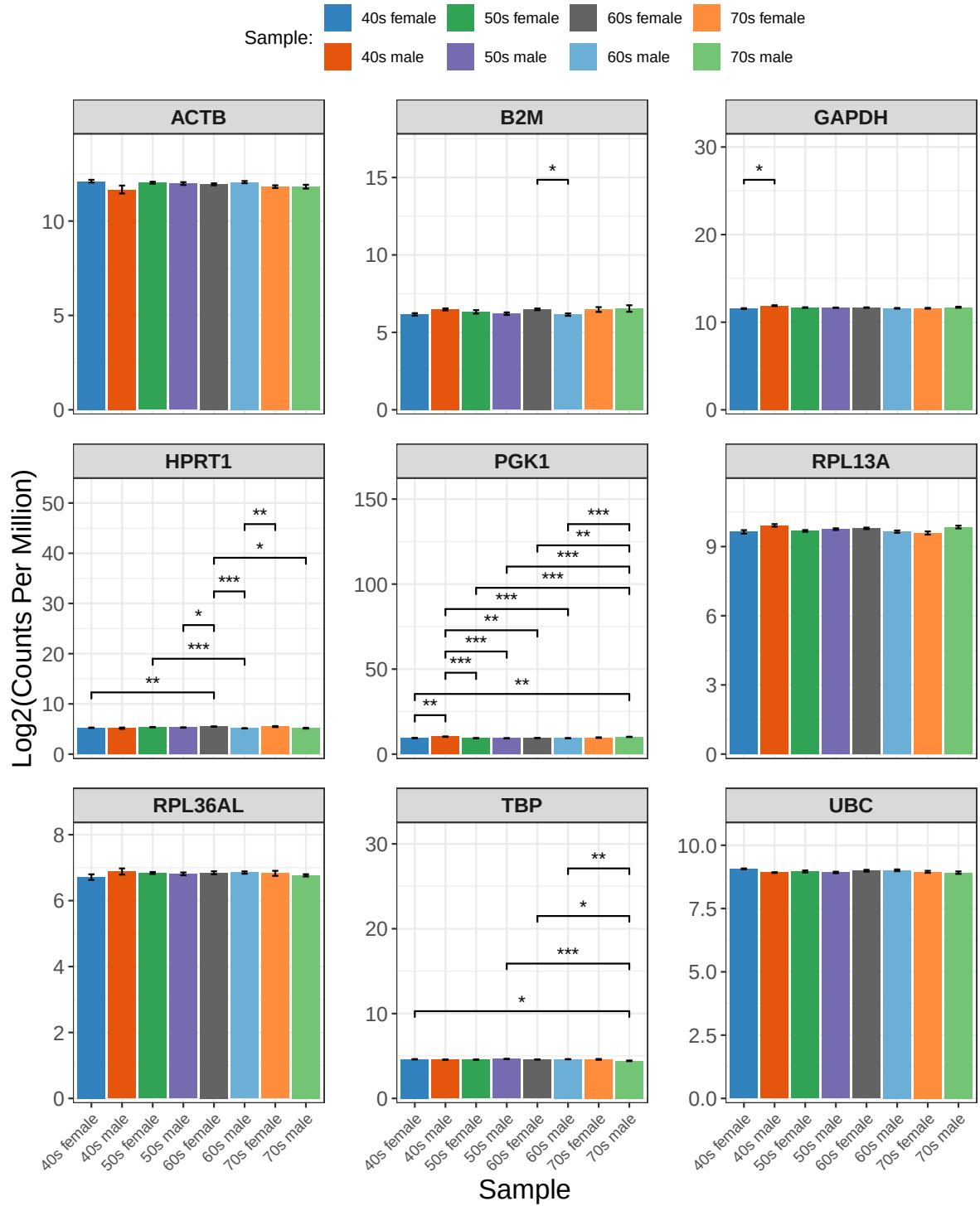


Figure 14: Expression of Housekeeping Genes

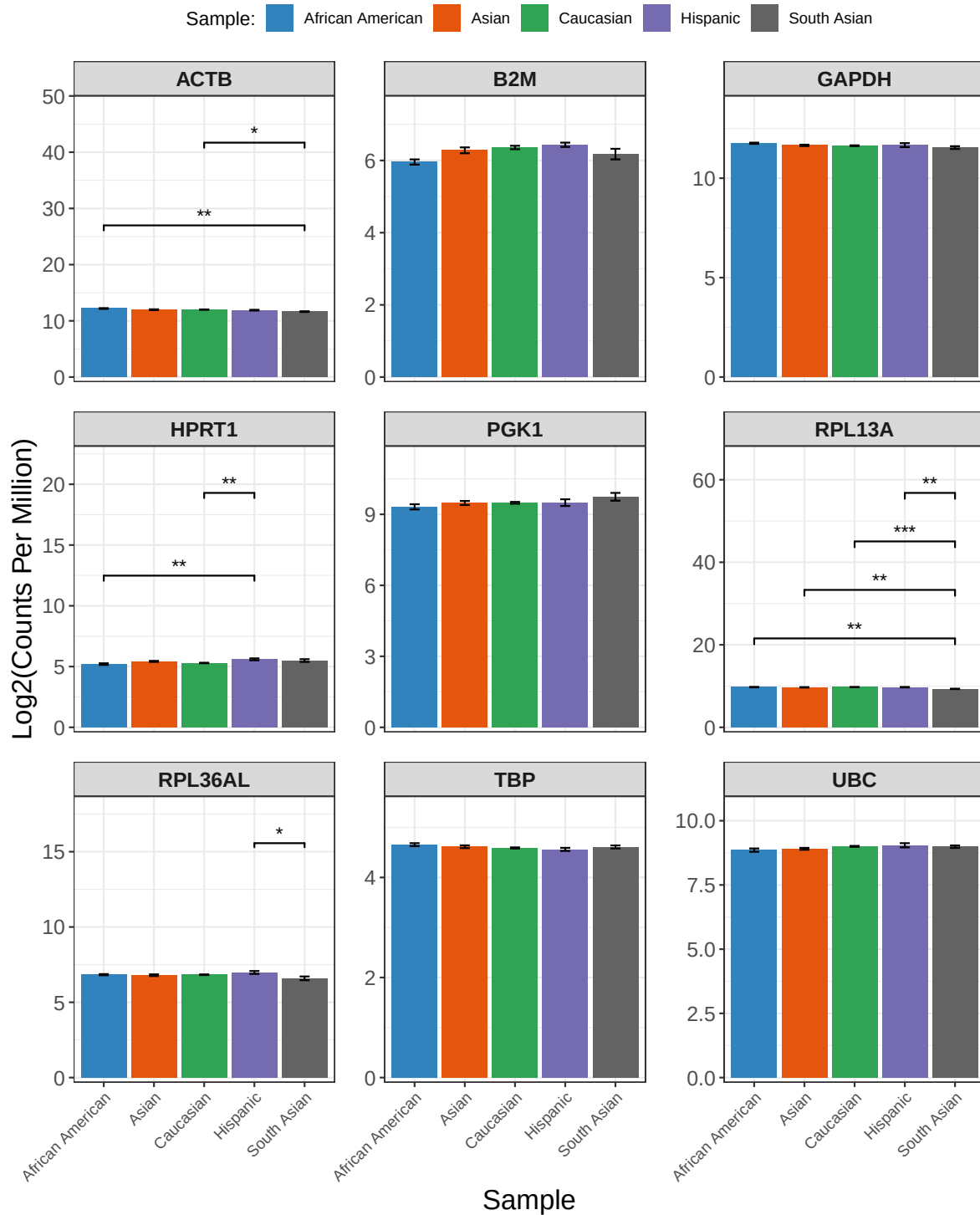


Figure 15: Expression of Housekeeping Genes

2	Residuals	284	6.59	0.0232	NA	NA	TBP	NA
3	group_var	7	2.50	0.357	1.93	6.55e-2	ACTB	0.0842
4	Residuals	284	52.6	0.185	NA	NA	ACTB	NA
5	group_var	7	11.7	1.67	6.29	7.05e-7	PGK1	0.00000317
6	Residuals	284	75.4	0.266	NA	NA	PGK1	NA
7	group_var	7	5.17	0.738	7.93	8.63e-9	HPRT1	0.0000000776
8	Residuals	284	26.4	0.0931	NA	NA	HPRT1	NA
9	group_var	7	1.06	0.152	2.21	3.32e-2	GAPDH	0.0499
10	Residuals	284	19.5	0.0687	NA	NA	GAPDH	NA
11	group_var	7	0.513	0.0733	1.11	3.56e-1	UBC	0.401
12	Residuals	284	18.7	0.0660	NA	NA	UBC	NA
13	group_var	7	0.532	0.0760	0.761	6.20e-1	RPL36AL	0.620
14	Residuals	284	28.3	0.0998	NA	NA	RPL36AL	NA
15	group_var	7	6.07	0.868	2.23	3.18e-2	B2M	0.0499
16	Residuals	284	110.	0.389	NA	NA	B2M	NA
17	group_var	7	1.75	0.250	2.54	1.51e-2	RPL13A	0.0340
18	Residuals	284	28.0	0.0985	NA	NA	RPL13A	NA

A tibble: 252 x 7

Groups: Gene [9]

	diff	lwr	upr	p_value	Gene	comparison	p_adj
	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<chr>	<dbl>
1	-0.0412	-0.226	0.143	0.997	TBP	40s male-40s female	1.000
2	-0.0393	-0.141	0.0627	0.938	TBP	50s female-40s female	1.000
3	0.0402	-0.0724	0.153	0.959	TBP	50s male-40s female	1.000
4	-0.0230	-0.123	0.0767	0.997	TBP	60s female-40s female	1.000
5	0.00760	-0.0972	0.112	1.000	TBP	60s male-40s female	1.000
6	-0.0131	-0.163	0.137	1.000	TBP	70s female-40s female	1.000
7	-0.180	-0.343	-0.0168	0.0193	TBP	70s male-40s female	0.180
8	0.00187	-0.173	0.176	1.000	TBP	50s female-40s male	1.000
9	0.0813	-0.0996	0.262	0.869	TBP	50s male-40s male	1.000
10	0.0181	-0.155	0.191	1.000	TBP	60s female-40s male	1.000

i 242 more rows

Discussion

Conclusion

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